DeFeng Zhou

defengzhou@miami.edu · 7866436012 · Elapsedf · elapsedf.github.io · Defeng Zhou

Education

University of Miami, Coral Gables, FL, USA — Sept. 2025 – Current

PhD in Department of Electrical and Computer Engineering
Supervisor: Prof. Mingzhe Chen (Director of WINS Lab)
GPA (Major Course): 3.9/4.0

Sun Yat-sen University, Guangdong, China — Sep 2021 – Jun 2025

B.E. in College of Intelligent Systems Engineering
Supervisor: Prof. Shimin Gong
GPA (Major Course): 3.8/4.0 (top 5% of majors in 2021-2022)

Research Interests

Wireless AI, semantic communications, machine learning, secure wireless transmission, unmanned aerial vehicles/drones

Selected Publication

D. Zhou, D. Wei, Y. Hu, S. Li and M. Chen, "Semantic Communication Performance Optimization with Channel and Content Preference Feedback," accepted by *2026 IEEE International Conference on Communications* (***2026 ICC***)

D. Zhou, S. Gong, L. Li, B. Gu and M. Guizani, "Deep Reinforcement Learning for IRS-assisted Secure NOMA Transmissions Against Eavesdroppers," accepted by *2024 International Wireless Communications and Mobile Computing* (***2024 IWCMC***)

D. Zhou, L. Li, S. Gong, B. Gu, G. Chen, D. Niyato, "Learning Simultaneous Information and Jamming Beamforming for Hybrid IRS-assisted Secure NOMA Transmissions", accepted by *IEEE Transactions on Communications* (***TCOM***)

Research Experience

WINS Lab, University of Miami — Aug 2025 – Present

Research Assistant Advisor: Mingzhe Chen

Topic: HD map-conditioned visual token pruning for autonomous driving VLMs (MapNeSt)

- Proposed MapNeSt, an HD-map-guided visual token pruning framework for autonomous driving VLMs, exploiting HD map priors to identify spatially redundant tokens and reduce visual context by up to 75% without degrading task performance.

Topic: Optimization of semantic communication with limited feedback

- Developed semantic communication optimization frameworks that jointly leverage channel state information and content/user preference feedback for resource allocation and feedback design.
- Extended to multi-user settings via distributed learning and multi-agent reinforcement learning to support scalable control in dynamic networks.

WINS Lab, Sun Yat-sen University — Jul 2023 – Apr 2025

Research Intern Advisor: Shimin Gong

Topic: Leverage NOMA and RIS for secure wireless transmission

- Designed a DRL-based control policy for passive RIS-assisted secure NOMA to jointly optimize transmission variables (e.g., power allocation, beamforming, and RIS phase shifts).
- Generalized passive RIS to hybrid RIS under practical energy constraints, and proposed lightweight learning-based alternatives to alternating-optimization pipelines.

YC Lab, Sun Yat-sen University Sep 2022 – Nov 2023

Research Intern Advisor: Calvin Yu-Chian Chen (Now Director of AI for Science at PKU)

Topic: Computer vision and natural language processing

- Built an image deraining model based on detail scaling and texture extraction to improve robustness across diverse rain patterns.
- Developed a voiceprint recognition pipeline using Wav2Vec features; improved multi-class speaker identification and validated generalization across public and private datasets.

♡ Honors and Awards

Silver Prize, 13th MathorCup Mathematical Modeling Challenge for Colleges and Universities 2023

The Third Prize Scholarship (Top 25%), Sun Yat-sen University 2022-2023 Academic Scholarship 2023

Interdisciplinary Talent Award (**Only 6 student in the whole college**), Sun Yat-sen University Intelligent Medical Interdisciplinary Talent Training Fund 2023

First Prize, 12th Asia and Pacific Mathematical Contest in Modeling (Also Best Programming Award and Best New Media Award) 2022

Ethics Award, Sun Yat-sen University 2021-2022 Specialized Scholarships 2022

The First Prize Scholarship (**Top 5%**), Sun Yat-sen University 2021-2022 Academic Scholarship 2022

Skills

- Programming: Python, C/C++, Matlab, LaTeX, HTML, R
- Framework: PyTorch, Gym, scikit-learn
- Devtools: Git, Linux
- Language: Chinese (Native Speaker), English (IELTS: 6.5, CET-6)

Academic Service

- Reviewer of *IEEE Transactions on Mobile Computing*, *IEEE Transactions on Communications*, *IEEE Transactions on Green Communications and Networking*, *Neural Computing and Applications*.
- Reviewer of *IEEE WF-IoT*, *IEEE Wireless Communications Letters*, *IEEE International Conference on Communications*.
- Google Scholar: https://scholar.google.com.hk/citations?hl=zh-CN&user=6B91xcQAAAAJ